Prayas JEE 2.0 (2024)

Quadratic Equations

DPP-05

1. If the equations $ax^2 + bx + c = 0$ and $x^2 + x + 1 = 0$ has one common root then $a : b : c$ is equal to
(1) 1 : 1 : 1 (2) 1 : 2 : 3
(3) 2 : 3 : 1 (4) 3 : 2 : 1

2. If the equation $px^2 + 2qx + r = 0$ and $px^2 + 2rx + q = 0$ $(q \neq r)$ have a common root, then $p + 4q + 4r$ equals
(1) 0 (2) 1
(3) 2 (4) –2

3. If the quadratic equation $3x^2 + ax + 1 = 0$ and $2x^2 + bx + 1 = 0$, have a common real root, then the value of the expression $5ab - 2a^2 - 3b^2$ is
(1) 0 (2) 1
(3) –1 (4) none of these

4. If the equation $x^2 - (p + q)x + 4 = 0$; $x^2 - 2qx + 8 = 0$ and $x^2 - (2p + q)x + 12 = 0$ have exactly one common root then which of the following is(are) CORRECT?
(1) $p = 2$
(2) $q = 3$
(3) Common root is 4
(4) Sum of all uncommon root is 6

5. The number of integral values of m for which the quadratic expression, $(1 + 2m)x^2 - 2(1 + 3m)x + 4(1 + m)$, $x \in R$, is always positive, is :
(1) 7 (2) 8
(3) 3 (4) 6

6. The integer 'k', for which the inequality $x^2 - 2(3k - 1)x + 8k^2 - 7 > 0$ is valid for every x in R, is
(1) 4 (2) 2
(3) 3 (4) 0

7. Consider $y = \dfrac{2x}{1+x^2}$, where x is real, then the range of expression $y^2 + y - 2$ is
(1) [–1, 1] (2) [0, 1]
(3) [–9/4, 0] (4) [–9/4, 1]

8. If $a > b > c > 0$ and $(a + b - 2c)x^2 + (a + c - 2b)x + b + c - 2a = 0$, then
(1) one root of the equations is negative
(2) If one root lies between –1 and 0 then $a + c - 2b < 0$
(3) If one root is less than –1 then $a + c - 2b > 0$
(4) the two roots cannot have same sign

9. a and b are distinct real numbers, equation $x^2 - 2ax + ab = 0$ has both roots positive then which of the following are true?
(1) $0 < b/a < 1$
(2) one root lies between 0 and b
(3) one of the roots lies between $2a - b$ and $2a$
(4) both the roots are real and distinct

10. If $a > b > c > 0$ and $(a + b - 2c)x^2 + (a + c - 2b)x + b + c - 2a = 0$ has one root in the interval (–1, 0) then **[Challenger]**
(1) roots of $ax^2 + 2bx + c = 0$ are real and negative
(2) roots of $cx^2 + 2ax + b = 0$ are real and negative
(3) roots of $bx^2 + 2cx + a = 0$ are real and negative
(4) none of these

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (1)
2. (1)
3. (2)
4. (1, 2, 3, 4)
5. (1)
6. (3)
7. (3)
8. (1, 2, 3, 4)
9. (1, 2, 3, 4)
10. (1, 2)